

EV Range Prediction — Competition Technical Report

1. Objective

Predict official EV range from static specifications using a leakage-safe regression workflow.

2. Data & Cleaning

The supplied dataset contains 478 EV models and 22 original fields. Cargo text values were parsed; missing values were handled appropriately; constant battery_type, metadata source_url and near-unique model were excluded. efficiency_wh_per_km was excluded from the final predictor.

3. Feature Engineering

Created footprint_m2, volume_proxy_m3, battery_per_torque, battery_per_footprint and performance_index. These features do not use the target or prohibited efficiency field.

4. Model Selection

CatBoost (Final) gave the strongest holdout performance. CatBoost is appropriate for the small mixed numerical/categorical dataset.

5. Validation

Five-fold CV mean: MAE 12.995 km, RMSE 18.244 km, R² 0.9679.

6. Deployment

The final package includes a reusable pipeline artifact and Streamlit interactive tester for live judge input.

7. Limitations

This is a specification-based estimate, not a real-time range estimator. Trip-level variables such as weather, traffic, HVAC, State of Charge and State of Health are absent.

Model Comparison

Model	MAE	RMSE	R ²
CatBoost (Final)	11.77	15.10	0.9784
Ridge Regression	15.22	18.53	0.9676
Random Forest	14.14	19.10	0.9655
Gradient Boosting	16.14	20.69	0.9596